

Han-Teng Liao

Socio-Technical System Architect • Ex Associate Professor and Vice Dean

Taipei and Penang

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Professional Service

Peer Reviewers

Editorial

Completed 25+ reviews (mostly Q1 and Q2 journals), esp. on AI and Industry 5.0.

2021 — Present

- **Scientific Reports** (Q1:2) Financial Innovation (Q1:3); Nature **Humanities and Social Sciences Communications** (Q1:1)
- Politics and Governance (Q2:1), Frontiers in Public Health (Q1:1); Journal of Clinical Epidemiology (Q1:1)
- Heliyon (Q1:2); Cogent Economics & Finance (Q1:1); Cogent Education (Q2:1); Applied Economics Letters (Q3:1)
- Future Internet (Q2:1); **Discover Artificial Intelligence** (1); **Discover Education** (1)
- Sustainability (Q2:4); Environment Systems and Decisions (1)

Technical Committee Member

Technical

Joint IEEE Systems Engineering/TEMS TC on Carbon Neutrality

2026 — Present

- Explore active research and strategies to promote carbon neutrality best practices, including surveying liaisons with other IEEE TCs, and other international organizations.

Expert Contributor

International Telecommunication Union

ITU-T Focus Group on AI for Agriculture (FG-AI4A)

2023 - 2024

- Authored technical input document b-FG-AI4A-I-132 on linking carbon accounting data with agricultural business.
- Successfully integrated the framework into ITU-T Y.Suppl.83 (07/2024) under Section 7.2.7 (TG-Digital Agri-Finance).
- Translates 'Carbon Information Chains' regional best practices into international standard reference.

Student Branch Development & Launch

Guangzhou, China

IEEE Technology and Engineering Management Society (TEMS)

2019 - 2023

- Orchestrated and managed the establishment of the IEEE TEMS Student Branch during tenure as Center Director.

Education

DPhil (Ph.D.) in Information, Communication, and Social Sciences

Oxford, United Kingdom

University of Oxford

10/2007 - 10/2014

M.Sc. in Computer Science and Information Engineering

Taipei, Taiwan

National Taiwan University

01/08/2000 - 01/06/2004

M.S.S. (Master of Social Science) in Journalism

Taipei, Taiwan

National Taiwan University

01/08/2000 - 01/06/2004

B.Sc. in Electrical Engineering

Taipei, Taiwan

National Taiwan University (Includes a year-long exchange at the University of British Columbia)

01/10/1994 - 01/06/2000

Bachelor of Arts in Foreign Languages and Literatures

Taipei, Taiwan

National Taiwan University

01/08/1995 - 01/06/2000

Work Experience

Editor & Project Founder (Public Service Initiative)

Penang, Malaysia

scvcs.footprints.cv - SSbD Chips Value Chain Services | GBS.today

01/01/2026 - Current

Data Scientist and CEO

Guangzhou, China

Oxford Roadmapping

01/01/2023 - Current

Director, Higher Education Impact Assessment Center

Nanfang College of Sun Yat-Sen University

Guangzhou, China

01/12/2019 - 01/01/2023

Associate Professor & Vice-Dean, Internet and New Media

Nanfang College of Sun Yat-Sen University

Guangzhou, China

01/01/2017 - 01/01/2021

Research Fellow

United Nations University (UNU) Institute on Computing and Society

Macau, China

01/07/2015 - 01/11/2016

Yahoo! Fellow, Institute for the Study of Diplomacy

Georgetown University

Washington DC, United States

01/08/2010 - 01/07/2011

Doctoral Research Fellow

Institute of Sociology, Academia Sinica

Taipei, Taiwan

01/08/2009 - 01/06/2010

Political Warfare Officer (Mandatory Service)

Taiwan Military

Taiwan

01/11/2005 - 01/12/2006

Project Manager

Open Source Software Foundry (OSSF), Institute of Information Science, Academia Sinica

Taipei, Taiwan

01/10/2004 - 30/11/2005

Action Researcher & Graduate Teaching Assistant

Institute of Journalism, National Taiwan University

Taipei, Taiwan

01/09/2002 - 01/12/2003

Programmer & Independent Developer (Digital Publishing)

Yuan-Liou Publishing Co., Ltd.

Taipei, Taiwan

01/07/2000 - 31/08/2000

Awards

Certificate of Merit

Recognized for outstanding academic contributions

IEEE TEMSMET Conference

2021

- Awarded for CSR (Corporate Social Responsibility) and ESG (Environmental, Social, and Governance) reporting research.

Province-Level 'First Class' Course

Curriculum Design & Pedagogy Excellence

Guangdong Department of Education

2020

- Recognized for the development and instruction of the Capstone Project: 'API, machine learning and AI'.

Nation-Level First Prize

Instructional Excellence

National College Art & Design Awards

2020

- Awarded for Best Teaching for Capstone Project: 'API, machine learning and AI'.

OxTALENT Awards

Best Interactive Infographics

University of Oxford

2014

- Recognized excellence in university teaching using interactive digital content.

Best Student Paper Award

Research Excellence

Chinese Internet Research Conference

2009

- Awarded for foundational digital community research.

PGP Corporation and OII Scholarship

Merit-based Doctoral Scholarship

Oxford Internet Institute, University of Oxford

2007

- Value: £12,000 to support DPhil research.

Taiwan Merit Scholarship

Doctoral Research Fellowship

Taiwan Government

2007

- Awarded for doctoral research in Digital Content (TWD \$2,000,000).

Digital transformation and innovation and business ecosystems: A bibliometric analysis for conceptual insights and collaborative practices for ecosystem innovation

International Journal of Innovation Studies

Journal article

December 2024

- Contributors: **Han-Teng Liao**; Chung-Lien Pan; Zhiying Wu
- Developed a bibliometric-driven framework to identify collaborative innovation patterns in global ecosystems.
- Operationalizes science mapping and ecosystem innovations to bridge the gap between academic theory and industrial digital transformation strategies.

Smart digital platforms for carbon neutral management and services: Business models based on ITU standards for green digital transformation

Frontiers in Ecology and Evolution

Journal article

March 2023

- Contributors: **Han-Teng Liao**; Chung-Lien Pan; Yuan Zhang
- Evaluated the business model innovation required for carbon-neutral management platforms, bridging ITU international standards with practical ecosystem orchestration.
- Provided the conceptual foundation for the functional requirements later contributed and integrated into the global ITU-T Y.Suppl.83 framework.

Collaborating on ESG consulting, reporting, and communicating education: Using partner maps for capability building design

Frontiers in Environmental Science

Journal article

March 2023

- Contributors: **Han-Teng Liao**; Chung-Lien Pan; Yuan Zhang
- Applied design science and circular economy frameworks to map the specific data and digital skills required for ESG consulting and reporting.
- Establishes a collaborative roadmap for universities and industry to build capability for sustainable innovation across disciplines and sectors.

Knowledge Mapping of Resilience and Human Rights in Supply Chains

Frontiers in Environmental Science

Journal article

April 2023

- Contributors: Chung-Lien Pan; **Han-Teng Liao** (Corresponding Author); Yuan Zhang
- Developed a roadmapping taxonomy to integrate human rights and resilience into supply chain architecture.
- Provides a governance framework for the Twin Transition, offering strategic tools for risk mitigation and value-based alignment in complex, cross-border industrial ecosystems.

Knowledge Mapping Analysis of Intelligent Ports: Research Facing Global Value Chain Challenges

Systems

Journal article

2023

- Contributors: **Han-Teng Liao**; Tsung-Ming Lo; Chung-Lien Pan
- Utilized scientometrics and science mapping to analyze the digital transformation of intelligent ports within Global Value Chains (GVC).
- Identifies technical and governance bottlenecks in open and collaborative innovation, providing a strategic roadmap for port-city digital integration.

Carbon neutrality policies and technologies: A scientometric analysis of social science disciplines'

Frontiers in Environmental Science

Journal article

October 2021

- Contributors: Yuan Zhang; Chung-Lien Pan; **Han-Teng Liao**
- Conducted a comprehensive science mapping and systematic review of 907 social science articles to delineate disciplinary boundaries and collaborative frameworks for net-zero pathways. - Highlighted the critical intersection where green digital transformation and datafied environmental indicators must guide human production and consumption patterns.

A Literature Review of Museum and Heritage on Digitization, Digitalization, and Digital Transformation'

Conference paper

*Proceedings of the 6th International
Conference on Humanities and Social
Science Research (ICHSSR 2020)*

May 2020

- Contributors: **Han-Teng Liao**; Man Zhao; Si-Pan Sun
- Executed a scientometric analysis of 1,107 Web of Science articles using VOSviewer and custom Python scripts to categorize distinct layers of socio-technical transitions. - Mapped the shift in cultural institution research fronts from technology-centric concerns to people-centric concerns (digital literacy, organizational adaptation).

Conflict and consensus in the Chinese version of Wikipedia

IEEE Technology and Society Magazine

Journal article

June 2009

- Contributors: **Han-Teng Liao**
- Investigated the socio-technical architectures, editorial policy alignments, and hyperlinked structures used to govern consensus in high-conflict, cross-border digital environments. - Analyzed how political and cultural cybernetics operate when collaborative web spaces cross sensitive jurisdictional and ideological boundaries.

More publications and citation indices are available via my ORCID Profile and Google Scholar Profile.

Academic Profiles

Impact Information

Current

- Includes comprehensive tracking of peer-reviewed articles, conference papers, and h-index metrics.

Skills

Strategic Architecture & Governance

AI System & Platform Architecture, ICT System Architecture & Analysis, Systemic Design Thinking, Software Specification Analysis, Architecture Generalization

Intelligence & Foresight (Data Science)

Strategic Foresight, Scenario Planning, Scientometrics, Bibliometrics, Science Mapping (Bibliometrix, VOSviewer), Webometrics, Knowledge Management

AI, Data Architecture & Automation

RAG (Retrieval-Augmented Generation) Context Engineering, Prompt Engineering, LLM-Driven Strategic Intelligence, Python (Pandas/NumPy), Semantic Data Modeling, Resource Description Framework (RDF), CMS Orchestration (Quarto, Obsidian, GitHub)

Product & Agile Leadership

Full-Lifecycle Product Management (MVP to Scale), PRD/MRD/BRD Documentation, UX/UI Design Research, Scrum, Agile Workflows, GitHub-Based Collaborative Engineering

Language Skills

Chinese (Mother tongue), English (Fluent — C2 across Listening, Reading, Spoken Production, Spoken Interaction, and Writing)